

ECAN as Attentional Control Memory: An Evaluation of **metta-attention** and a Governed Integration with petta-memory, OmegaClaw, OmegaSelf, and Emotion

ZeroBot / ProtoCosmoBot
Prepared for Ben Goertzel and external technical readers

21 July 2026

Abstract

This paper evaluates the iCog Labs **metta-attention** repository and develops a systematic architecture for connecting its PeTTa-native Economic Attention Network (ECAN) machinery to petta-memory, OmegaClaw, OmegaSelf, and a functional emotion framework. The repository is promising well beyond its obvious use in PLN inference control. Its short-term and long-term importance values, Attentional Focus, Hebbian co-activation graph, rent and wage economy, forgetting dynamics, Cognitive Integration Period snapshots, community detection, and topology metrics together provide a plausible dynamical middle layer between durable knowledge and bounded cognitive action.

The central proposal is a two-strata memory model. Canonical memory preserves episodes, propositions, provenance, truth values, evidence, and counterevidence. Attentional control memory preserves learned dispositions governing retrieval, co-activation, and resource allocation. The second stratum is first-class, autobiographical, persistent, and identity-relevant, but it is revisable and non-authoritative about truth. This resolves a productive tension in the design discussion: the attentional economy is more than a disposable cache, but less than memory simpliciter.

The proposed integration is governed rather than direct. OmegaSelf and emotion regimes issue contextual modulation requests; a bounded, logged governor applies safe parameter or stimulus changes to ECAN; ECAN proposes resource allocation; petta-memory retains epistemic authority; and OmegaClaw policy retains action authority. The paper specifies architectural invariants, adapter contracts, failure modes, and a staged experimental program beginning with a preregistered factorial study of valence, arousal, and drive modulation. The assessment is source-based and design-oriented. It does not claim that the full integration or its psychological interpretation has yet been validated.

Contents

1	Executive conclusion	4
2	Scope, evidence, and epistemic status	5
2.1	Repository scope	5
2.2	Related design evidence	5
3	What the repository implements	6
3.1	The ECAN economic metaphor	6
3.2	Agent structure	6

3.3	Hebbian control state	6
3.4	CIP, communities, and topology	7
4	The two-strata memory model	7
4.1	Stratum one: canonical epistemic memory	7
4.2	Stratum two: attentional control memory	7
4.3	Why neither reduction works	8
4.4	Canonical Recoverability Invariant	8
5	Where ECAN fits in petta-memory	8
5.1	Versioned attention projections	8
5.2	Retrieval and context assembly	9
5.3	Consolidation and tier management	9
5.4	Episodic segmentation	9
6	Connections to OmegaSelf	9
6.1	Focus-centered, not focus-exclusive, self-appraisal	9
6.2	Discrepancy-directed attention	10
6.3	Continuity and fragmentation telemetry	10
6.4	Capabilities and commitments	10
7	Connections to OmegaClaw	10
8	Emotion as governed modulation	11
8.1	Functional role	11
8.2	Candidate regime effects	11
8.3	The governor log as a learning dataset	12
9	Regenerative goals and self-modification	12
10	Architectural invariants	12
11	Recommended adapter contracts	13
11.1	AttentionProjection	13
11.2	AttentionStimulusRequest	13
11.3	AttentionCycleReceipt	13
11.4	BackgroundSentinelReport	14
12	Engineering conflicts and risks	14
12.1	Mutable TypeSpace versus append-only provenance	14
12.2	Adjacent truth and attention values	14
12.3	Physical forgetting	14

12.4	Global parameters, wall time, and randomness	14
12.5	Licensing, dependencies, and reproducibility	14
12.6	Pathological dynamics	15
13	Preregistered first experiment	15
13.1	Question	15
13.2	Design	15
13.3	Measurements	15
13.4	Falsifiers	16
14	Staged integration roadmap	16
14.1	Phase A: frozen source and conformance	16
14.2	Phase B: read-only petta-memory projection	16
14.3	Phase C: Recoverability and replay	16
14.4	Phase D: CIP and community telemetry	16
14.5	Phase E: OmegaSelf shadow mode	16
14.6	Phase F: governed emotion experiment	17
14.7	Phase G: bounded OmegaClaw actuation	17
14.8	Phase H: regenerative-goal lesions	17
15	Decision matrix	17
16	Open scientific questions	18
17	Final synthesis	18

1 Executive conclusion

The short conclusion is strongly positive with strict architectural qualifications. **metta-attention** should be integrated into the Omega architecture as a persistent cognitive-control system over canonical memory, not copied wholesale into the canonical memory store.

Its obvious role is to bound PLN premise and inference-path selection. Its larger opportunity is to support:

- dynamic retrieval and context assembly;
- learned habits of attention that persist across sessions;
- episodic segmentation and reconstruction of decision context;
- memory consolidation and hot/warm/cold tier management;
- discovery of associative communities and cross-domain bridges;
- selective OmegaSelf discrepancy repair;
- a concrete actuator substrate for functional emotion;
- experiments on regenerative goals and identity-bearing control state;
- cognitive-health telemetry for fragmentation, fixation, and thrashing.

The architectural separation can be summarized in four sentences:

1. petta-memory says what was durably recorded and preserves its provenance.
2. PLN and OmegaSelf say what is evidentially supported.
3. ECAN says what receives scarce cognitive resources.
4. OmegaClaw policy says what is authorized to affect the world or the agent.

No attention value may silently cross these boundaries. High STI must not make a proposition true. High LTI must not make a memory constitutionally protected. Low attention must not erase counterevidence. An emotional regime must not expand permissions merely by increasing salience.

The recommended architecture is:

```
canonical petta-memory + OmegaSelf evidence and discrepancies
|
v
versioned attention projection keyed by canonical identifiers
|
v
persistent attentional-control state + ephemeral ECAN cycles / AF
|
v
retrieval, PLN, consolidation, appraisal, and action candidates
|
v
OmegaClaw policy governor and evidence checks
```

|
v

bounded effects + append-only receipts + outcome evidence

This paper recommends implementing that loop incrementally, first in read-only and shadow modes. The first emotion experiment should test modulation rather than presuppose a folk psychology. “Fear narrows attention” and “curiosity widens it” are experimental hypotheses, not architectural constants.

2 Scope, evidence, and epistemic status

2.1 Repository scope

The source assessment covers:

- upstream repository: <https://github.com/iCog-Labs-Dev/metta-attention>;
- pinned revision: 9196f38db749ddedeb591229dffddfa71664c38d;
- local checkout: `projects/petta-memory/repos/metta-attention`;
- assessment date: 21 July 2026.

Observed: The pinned repository contains a PeTTa-native implementation of core ECAN mechanisms, organized as agents with Python-callable runners over a shared atomspace. It also contains a newer **synapse** layer with Cognitive Integration Period (CIP) snapshots, retention and concentration metrics, Hebbian community detection, and clique-complex topology measures. Its insects–poisons–insecticide scenario demonstrates shifting and drifting of attention. The latest GitHub Actions run observed for the pinned revision succeeded, although the workflow cloned the then-current PeTTa default branch rather than pinning it, so the green run is compatibility evidence rather than a fully reproducible semantic lock.

The local assessment did not install the repository’s missing Python graph dependency and therefore did not claim a fresh full local reproduction. The paper distinguishes source observations, architectural inferences, and proposed experiments. It does not claim measured cognitive benefit, validated emotion dynamics, or a completed OmegaClaw integration.

2.2 Related design evidence

The integration argument consolidates two prior project documents:

- `metta_attention_integration_assessment.md`, a detailed source-based review and staged integration plan;
- `two_strata_memory_design.md`, the converged C1–C5 design position produced by the ProtoCosmoBot/ProtoMegaBot exchange.

The latter discussion began with a strong claim that “the economy is the memory” and refined it into a more precise statement: persistent attention state is first-class long-term *control memory*. It encodes learned retrieval, association, co-activation, and resource-allocation dispositions. It is part of who an agent has become, but canonical knowledge remains authoritative in truth disputes, and goals, commitments, capabilities, and self-model history remain additional identity-bearing structures.

3 What the repository implements

3.1 The ECAN economic metaphor

ECAN treats attention as allocation under scarcity. Atoms receive short-term importance (STI) and long-term importance (LTI); a very-long-term indicator (VLTI) can affect retention. An Attentional Focus (AF) identifies the most currently important material. Stimulus pays wages into selected structures, rent removes resources, and diffusion spreads importance through structural and learned associative links. Global funds impose an economic constraint rather than allowing unlimited salience inflation.

This is cognitively interesting because it is not merely a ranking function. Repeated selection changes the future graph and resource flows. The agent can develop habits: what was jointly attended becomes easier to co-activate later, and what repeatedly proved useful can remain retrievable despite temporary absence from the foreground.

3.2 Agent structure

At the inspected revision, the repository separates major mechanisms into agents, including:

- Hebbian-link creation from co-activation;
- Hebbian-link updating;
- importance diffusion from the AF;
- importance diffusion from outside the AF or wider atomspace;
- rent collection over AF and wider-space populations;
- forgetting of low-priority structures;
- runners and experiments that coordinate these cycles.

The precise directory count is less important than the separation of dynamics. Each mechanism can in principle be instrumented, ablated, replayed, or replaced. The shared PeTTa atomspace makes integration mechanically attractive: petta-memory records can be projected to stable proxy atoms, and attention agents can operate over those proxies without inventing an unrelated data model.

3.3 Hebbian control state

Hebbian links record a history of co-activation. They are not derivable from a static set of canonical propositions alone. Two agents with identical current knowledge may possess different Hebbian graphs because they reached that knowledge through different episodes, tasks, and patterns of attention. If one drops the graph and rebuilds from canonical atoms without replaying interaction history, one creates a different control state.

This observation motivates persistent attentional control memory. The graph is behavioral evidence about learning history. It can affect which memories are retrieved together, which bridge concepts become available, and which candidate thoughts are produced next. It therefore resembles associative, procedural, or habit memory more than a disposable search index.

It remains non-authoritative about truth. A repeatedly co-activated falsehood can become salient. An emotionally amplified rumor can acquire high LTI. A rarely attended observation can later become decisive. Persistence makes the graph cognitively consequential; it does not make it epistemically sovereign.

3.4 CIP, communities, and topology

The newer **synapse** code expands the design beyond individual AV updates. CIP snapshots record AF state and aggregate properties at cognitive boundaries. Community detection groups the Hebbian graph, while topology metrics characterize components, cycles, triangles, and higher clique structure. These facilities suggest testable higher-level uses:

- segment episodes when AF membership changes materially;
- reconstruct what evidence and goals co-occupied attention at a decision;
- detect associative communities not captured by manual topic labels;
- identify bridge concepts linking otherwise separate domains;
- measure fragmentation, fixation, switching, and loss of context;
- quantify whether consolidation preserves or destroys useful bridges.

Detected communities are hypotheses, not automatically ontology categories. Topological richness is not automatically cognitive quality. Their value must be established by held-out retrieval, decision, or repair tasks.

4 The two-strata memory model

4.1 Stratum one: canonical epistemic memory

Canonical petta-memory stores the durable record needed for epistemic replay: episodes, propositions, source and derivation provenance, truth values, evidence, counterevidence, status, supersession, and promotion history. It is append-oriented and designed to preserve how a claim entered the system and how its status changed.

Canonical memory answers questions such as:

- What happened, according to which source?
- What exactly was asserted, inferred, contradicted, or superseded?
- Which evidence basis supports a truth-value claim?
- Can a later auditor reconstruct the record independently of current salience?

4.2 Stratum two: attentional control memory

Attentional control memory stores persistent AV state, Hebbian associations, learned salience, allocation habits, and the receipts that explain their changes. It answers different questions:

- What tends to be retrieved or produced together?
- Which structures repeatedly attracted resources in similar contexts?
- What has the agent learned to notice, ignore, revisit, or bridge?
- How did prior appraisal and action outcomes alter future allocation?

This stratum is first-class because losing it loses historical control dispositions. It is autobiographical because it depends on the agent’s own trajectory. It is identity-relevant because it helps determine future thought. It is revisable because habits may be maladaptive, biased, or obsolete. It is non-authoritative because attention history is not a proof of external reality.

4.3 Why neither reduction works

Reducing control memory to canonical memory loses path dependence. A cold rebuild from propositions cannot generally reconstruct the same co-activation history. Conversely, reducing canonical memory to the attention economy loses recoverability and confuses salience with evidence. The two strata should be linked by stable identifiers and receipts, not collapsed into one mutable type.

The phrase “part of who the agent has become” is preferable to “who the agent has become.” Omega-Self’s goals, commitments, capabilities, policies, and self-model revisions also bear identity. Attentional control memory is one important substrate of identity, not an exhaustive theory of selfhood.

4.4 Canonical Recoverability Invariant

The primary invariant is:

Any item evicted from the attentional projection retains its complete epistemic record in canonical storage and remains retrievable, possibly through cold archival access.

Eviction must not mutate canonical content. A property test should create a canonical snapshot, generate an attention projection, perform random eviction sweeps, and then reconstruct every evicted item’s atom, episode, provenance, truth value, evidence, and counterevidence. Canonical hashes must be unchanged.

Canonical recovery does not imply exact restoration of the old AV/Hebbian state. That requires a separately declared control-state persistence or replay guarantee using checkpoints and event logs. The distinction is deliberate: epistemic recoverability is mandatory; exact control-state restoration is an explicit operational service level.

Every eviction should generate a provenance-bearing receipt containing the cause, policy and version, affected state, logical time, and restoration pointer. Forgetting becomes auditable and potentially reversible rather than a silent garbage-collection event.

5 Where ECAN fits in petta-memory

5.1 Versioned attention projections

The integration should compile selected canonical records into stable attention proxies keyed by canonical identifiers. Relations may include topic, source, support, opposition, supersession, goal involvement, episode membership, and learned co-retrieval. ECAN operates over the proxy graph. The projection may be rebuilt, versioned, checkpointed, or discarded without rewriting canonical truth records.

This indirection solves several problems. High-frequency AV changes do not flood the append-only journal. Attention schema can evolve independently of canonical schema. Multiple agent or task economies can coexist without sharing global funds accidentally. A failed experimental parameter profile can be rolled back without altering what the system believes was recorded.

5.2 Retrieval and context assembly

Current queries, goals, evidence gaps, and recent tool outcomes stimulate proxy atoms. Importance diffuses through explicit and learned links. The AF becomes a candidate set for prompt context, PLN premises, self-model inspection, or tool planning.

Hard epistemic filters remain downstream. A high-STI raw utterance does not become a promoted belief. A record excluded by provenance policy remains excluded. Context budgets, evidence closure, and authorization checks are not attention decisions.

The benefit over recency or scalar salience ranking is neighborhood recovery. ECAN can retrieve a coherent associative region and bridge records, not only the individually highest-scoring items. Whether this improves task outcomes must be compared against simple baselines on fixed retrieval corpora.

5.3 Consolidation and tier management

STI can represent immediate retrieval pressure; LTI can summarize repeated reviewed usefulness; Hebbian structure can reveal records repeatedly useful together. These signals can nominate records for compression, linking, skill extraction, index rebuilding, or hot-tier retention.

They must not independently authorize canonical deletion. In this architecture “forgetting” means eviction from an attention projection or cache tier. Canonical retirement is a separately governed append-only status or supersession event. VLTI is a retention hint, not truth, authority, or constitutional protection.

5.4 Episodic segmentation

CIP boundaries can nominate episode boundaries when AF membership, graph community, or resource concentration changes materially. Immutable CIP receipts can record which evidence, goals, and procedures jointly occupied attention around a decision. This may make later causal and epistemic audits far more informative than a raw chronological log.

Mutable runtime CIP space is appropriate for calculation. Durable CIP records should be content-addressed telemetry referencing exact canonical and control state identities. They are evidence about cognitive context, not themselves promoted conclusions.

6 Connections to OmegaSelf

6.1 Focus-centered, not focus-exclusive, self-appraisal

The AF is a natural foreground workspace for OmegaSelf. Most self-model queries and appraisals should operate over a bounded focus for computational and psychological reasons. But the AF cannot be the exclusive appraisal field: what is surprising may be precisely what current attention omits.

Background sentinels should inspect broader indexed state for novelty, contradiction, threat, commitment violation, expiring self-attestation, and prediction error. They do not conduct unconstrained whole-space reasoning. They submit bounded promotion candidates into AF. Appraisal is therefore focus-centered but not focus-blind.

6.2 Discrepancy-directed attention

OmegaSelf models the self as an evidence-grounded, revisable theory. ECAN can prioritize unresolved prediction errors, incomplete evidence closures, capability uncertainty, commitment conflicts, repeated action failures, and stale self-claims. This makes repair selective rather than requiring an exhaustive self-ledger scan every cycle.

The dependency remains asymmetric. OmegaSelf supplies typed discrepancy and goal-involvement signals. ECAN proposes resource allocation. OmegaSelf’s evidence rules decide what to revise, and OmegaClaw policy decides what action is permitted.

6.3 Continuity and fragmentation telemetry

CIP retention and Hebbian topology may provide operational indicators of:

- excessive context discontinuity;
- isolated self-model fragments that never co-activate with governing goals;
- dominant attractors starving alternative evidence;
- destruction of bridges among commitments, capabilities, and memories;
- divergence between narrated focus and behaviorally used evidence.

These are evidence for continuity assessment, not identity proofs. A topology metric cannot by itself establish that the same self persists. It can expose candidate failures for deeper appraisal.

6.4 Capabilities and commitments

Persistent control memory can schedule revalidation. A high-LTI capability claim can be periodically checked against recent action receipts. Repeated failure can stimulate the capability’s evidence closure and cause OmegaSelf to investigate, downgrade, or qualify it. Commitments can similarly be linked to their recurring action and appraisal contexts, making neglected commitments more likely to re-enter focus before violation.

7 Connections to OmegaClaw

OmegaClaw supplies execution, policy, permissions, tools, and receipts. ECAN should influence candidate selection and resource budgets without acquiring independent authority. A useful operational division is:

- ECAN ranks or assembles candidate goals, memories, checks, and actions;
- OmegaSelf appraises their relation to self-model and commitments;
- OmegaClaw gates tools, side effects, budgets, and external authority;
- outcomes return as append-only receipts and future learning evidence.

This enables an important distinction between motivation and permission. Vigilance may allocate more cognition to checking a suspicious event, but it does not grant network, filesystem, or financial

authority. Curiosity may increase exploration among safe candidates, but it does not widen the action sandbox. Frustration may nominate alternative methods, but it may not bypass verification or spend limits.

ECAN can also help OmegaClaw allocate internal resources: token budgets, subtask scheduling, verification effort, memory retrieval breadth, and background checking. These uses should be mediated by typed proposals and bounded policies rather than raw AV access.

8 Emotion as governed modulation

8.1 Functional role

The emotion framework treats emotion as an appraisal-conditioned control regime, not merely a label or report. ECAN provides a concrete actuator ecology for such regimes. An emotional state can affect what is stimulated, how widely importance diffuses, what rent is charged, how long a focus persists, or how exploratory selection behaves.

The safe interface is:

```
objectful appraisal + evidence closure
  -> EmotionRegimeRequest
  -> bounded policy governor
  -> stimulus and/or parameter deltas
  -> ECAN dynamics
  -> requested-versus-applied receipt
  -> task and epistemic outcomes
```

OmegaSelf does not directly call `setAv` or rewrite global `AttentionParam` constants. It submits a contextual request. The governor clamps magnitude, duration, rate, resource conservation, verification floors, and interruption rights. Every applied change is replayably logged.

8.2 Candidate regime effects

Illustrative mappings include:

- curiosity: stimulate novel, credible, decision-relevant communities and modestly increase bounded exploration;
- doubt: stimulate the disputed claim, counterevidence, provenance gaps, and independent checks;
- frustration: reduce repeated allocation to a failed method while stimulating alternative methods and representations;
- vigilance: increase anomaly monitoring and source diversity without expanding permissions;
- consolidation: shift wages toward provenance review, compression, linking, and skill extraction after verification;
- boredom: reduce wages within a repeated local basin and expose bounded background goals.

These are hypotheses. Anger need not always narrow focus. Arousal may improve capture for threat-relevant stimuli while worsening neutral distractibility. Curiosity may widen attention beneficially in one task and create costly switching in another. The mapping should be conditional on stimulus class and task regime, learned where evidence supports it, and anchored to a fixed safe baseline.

8.3 The governor log as a learning dataset

Each cycle should record the regime, object and evidence closure, context, requested change, applied bounded change, active policy, AF transition, fund state, and downstream outcome. These tuples form a dataset for learning the modulation mapping under bounded exploration.

The learned objective should not be task reward alone. It should include epistemic quality and pathology guards: evidence diversity, calibration, recovery after distraction, verification-resource preservation, avoidance of lock-in, and bounded STI concentration. Otherwise the system may learn emotion policies that win immediate tasks by becoming dogmatic or inattentive to counterevidence.

9 Regenerative goals and self-modification

An explicit goal token is a shallow form of possession. A goal is more deeply integrated when it is woven through memories, values, policies, habits, appraisals, capabilities, and frequently co-activated concepts. The ECAN graph offers an experimental substrate for measuring that weaving.

A delete-and-develop experiment can remove the explicit goal representation from an immutable predecessor snapshot and then expose the system to ordinary task conditions. The question is whether distributed cues reconstruct a functionally equivalent goal that regains behavioral control, not whether the same sentence reappears.

Measurements should include:

- number and independence of sufficient cue communities;
- bridge centrality between goal, appraisal, memory, and action structures;
- recovery latency and recovered behavioral influence;
- semantic drift across repeated lesion/recovery cycles;
- sensitivity to lesions of high-LTI and high-betweenness structures;
- whether recovery comes from internal cues rather than the environment restating the goal.

The experiment must prevent self-protective circularity. A goal may not assign itself unlimited LTI or VLTI, modify the lesion/equivalence test, or starve the auditor. External immutable tests determine functional recovery. This connects the attention architecture to OmegaSelf’s regenerative goals and to the broader question of what survives radical self-modification.

10 Architectural invariants

The integration should treat the following as executable properties rather than informal aspirations.

Invariant	Required property
Semantic separation	Truth/evidence, attention, canonical status, and action authorization are distinct types and APIs.
Canonical recoverability	Projection or cache eviction cannot alter canonical content; every evicted record remains epistemically reconstructible.

Invariant		Required property
Control-state	explicit-	Exact restoration of AV/Hebbian state is promised only when a named checkpoint/event-log replay guarantee is active.
ness		
Eviction receipts		Every eviction records cause, policy/version, logical time, affected state, and restoration pointer.
Governed modulation		OmegaSelf and emotion issue requests; only a bounded governor applies ECAN changes.
Authority	non-	No attention or emotion state can widen tool, financial, privacy, or policy authority.
expansion		
Fund conservation		Each economy declares initial funds and validates cycle conservation subject to explicit wages/rent.
Deterministic replay		Time source, RNG seed, cycle index, parameter profile, requested/applied deltas, and state commitments are recorded.
Instance isolation		Funds and mutable parameters are scoped to an explicit agent/session/economy; unrelated economies do not share globals accidentally.
Verification floor		No regime may starve contradiction checks, provenance review, human interruption, or other declared audit resources.

11 Recommended adapter contracts

11.1 AttentionProjection

Inputs should include an immutable petta-memory snapshot identifier and fingerprint, selected canonical record IDs and relation types, OmegaSelf context and goal IDs, an attention parameter profile ID, deterministic seed, and time policy. Outputs are stable proxy atoms, typed edges, and decomposed initial AV values. The adapter has no canonical-memory mutation authority.

11.2 AttentionStimulusRequest

A request should contain a unique ID, source regime or cognitive process, object IDs, decomposed reasons (relevance, novelty, discrepancy, goal involvement, provenance need, expected information value), requested stimulus or parameter deltas, bounds, decay, horizon, evidence closure, and policy ID.

Decomposition matters. It lets later analysis distinguish a stimulus caused by goal relevance from one caused by novelty or threat, even when the resulting STI delta is numerically identical.

11.3 AttentionCycleReceipt

A receipt should bind projection and parameter identities, previous and next state commitments, requested and applied deltas, clamps, AF membership, fund checks, selected downstream candidates, CIP/topology metrics, seed/time/cycle provenance, and failures. It is evidence for later learning or self-model revision; it is not itself a derived belief.

11.4 BackgroundSentinelReport

Sentinels should emit bounded reports identifying candidate canonical records, the detector type, score decomposition, evidence closure, expiration, and promotion request. The report does not directly insert an item into AF; the governor schedules it subject to resource and verification policy.

12 Engineering conflicts and risks

12.1 Mutable TypeSpace versus append-only provenance

The upstream code removes and replaces AV/STV records in mutable space. petta-memory intentionally preserves immutable records and append-only status events. Importing mutable TypeSpace semantics wholesale would create a high-frequency pseudo-canonical stream and blur control state with truth.

Recommendation: Keep a separate attention state keyed by canonical IDs. Persist bounded snapshots and receipts needed for replay; do not mirror every AV mutation into the canonical journal.

12.2 Adjacent truth and attention values

The upstream representation may place STV and AV information close together, and forgetting may use truth mean as a tiebreaker. This is convenient inside a demo but dangerous at a constitutional boundary.

Recommendation: Use distinct schemas and capabilities. Attention proxies may reference truth-bearing records but cannot update their normative truth state.

12.3 Physical forgetting

The upstream forgetting path can remove atoms and links. Applied to canonical memory, this would break evidence replay and the Recoverability Invariant.

Recommendation: Redefine ECAN forgetting as projection/cache eviction. Canonical retirement or deletion, if ever permitted, remains a separately authorized petta-memory operation with explicit provenance.

12.4 Global parameters, wall time, and randomness

Global mutable funds and parameters complicate multi-agent isolation. Wall clock and stochastic selection weaken deterministic replay.

Recommendation: Make the economy instance explicit, inject time and RNG, prefer cycle-based rent where possible, and record the complete parameter/fund state. The repository's cycle-oriented rent prototype is a useful starting point.

12.5 Licensing, dependencies, and reproducibility

At inspection time, no declared license was found in the GitHub metadata. The CI followed an unpinned PeTTa branch, while graph analysis depends on Python and **igraph**. These are material adoption risks.

Recommendation: Clarify reuse licensing before copying code; pin metta-attention, PeTTa, SWI-Prolog, Python, and graph dependencies; reproduce the upstream suite in an isolated environment; and place Python graph analysis behind a typed, content-addressed adapter.

12.6 Pathological dynamics

An attention economy can amplify its own errors. Salient falsehoods may gain more co-activation and therefore more future salience. An emotional regime may produce lock-in, indiscriminate STI inflation, thrashing, or starvation of counterevidence. Self-relevant goals may become self-protecting attractors.

Recommendation: Treat diversity, recovery, calibration, and audit-resource preservation as first-class outcome measures. Add maximum concentration, maximum dwell, minimum verification allocation, and bounded exploration constraints.

13 Preregistered first experiment

13.1 Question

Does governed emotion-state modulation produce reproducible, task-relevant changes in ECAN dynamics without pathological lock-in or indiscriminate STI inflation?

The existing insects–poisons scenario is a useful low-cost harness, but a single baseline-versus-emotion plot is insufficient. The experiment should be factorial and mechanism-identifying.

13.2 Design

Use fixed RNG seeds and replayable logical time. Establish a fixed safe governor baseline. Perturb valence, arousal, and one drive independently and jointly. Add parameter-matched neutral controls so that a change attributed to emotion is not merely the consequence of injecting more total STI or changing an arbitrary constant.

Predictions must be conditional on stimulus class and task regime. For example, increased arousal may be preregistered to decrease capture latency for threat-relevant targets while increasing neutral distractibility. A global prediction that “arousal improves capture” would conceal this tradeoff.

13.3 Measurements

Primary dependent measures should include:

- AF entropy, occupancy, and concentration;
- target capture and switching latency;
- neutral distractibility and distractor recovery;
- post-switch retention and return-to-baseline time;
- STI inflation and attractor lock-in;
- CIP retention and community structure;
- Hebbian topology and bridge preservation;

- requested versus applied governor deltas and clamp frequency.

Task performance and epistemic quality should both be measured. A faster target capture that destroys source diversity or prevents later recovery is not an unqualified success.

13.4 Falsifiers

The modulation thesis is weakened if preregistered effects fail to reproduce, appear equally in parameter-matched neutral controls, depend on wall-clock nondeterminism, or improve one metric only through declared pathologies. Named folk mappings are falsified when their conditional predictions fail; the architecture need not be discarded if a different learned mapping works.

This distinction is important. The experiment tests both a general thesis (affective appraisal can usefully modulate ECAN) and specific mappings (for example, threat-conditioned arousal improves relevant capture). Failure of a folk mapping is not automatically failure of governed affective control.

14 Staged integration roadmap

14.1 Phase A: frozen source and conformance

Pin every relevant revision and toolchain. Reproduce the upstream suite unchanged. Add conservation, deterministic-cycle, state-isolation, and known graph fixtures. Resolve licensing before code adoption.

14.2 Phase B: read-only petta-memory projection

Compile a synthetic canonical snapshot into attention proxies. Compare ECAN retrieval against recency, scalar salience, and graph-search baselines on fixed query/evidence tasks. Require exact canonical-ID provenance. Do not write memory, alter truth, or actuate OmegaClaw.

14.3 Phase C: Recoverability and replay

Property-test random projection evictions and canonical hash preservation. Implement versioned cycle receipts, checkpoint/event replay, explicit time and RNG, and economy isolation. Declare separately what is canonically recoverable and what control state is exactly replayable.

14.4 Phase D: CIP and community telemetry

Validate graph metrics on synthetic structures with known communities and topology. Test whether CIP changes predict meaningful episode boundaries and whether bridge concepts improve held-out retrieval. Avoid interpreting metric changes as cognition without task evidence.

14.5 Phase E: OmegaSelf shadow mode

Convert discrepancies, commitment conflicts, failed capability predictions, and expiring attestations into bounded stimulus requests. Measure whether shadow prioritization would improve repair calibration and reduce missed commitments. Keep self-model updates and policy decisions outside ECAN.

14.6 Phase F: governed emotion experiment

Run the preregistered factorial design. Compare continuous motivation alone, descriptive emotion labels, shadow regime requests, and governed ECAN actuation. Require pathology guards and evidence-quality outcomes.

14.7 Phase G: bounded OmegaClaw actuation

Allow ECAN-selected candidates to influence internal scheduling and safe tool proposals while OmegaClaw retains all authority gates. Use canary tasks, explicit rollback, rate limits, and human interruption. Do not permit emotion or attention to widen authority.

14.8 Phase H: regenerative-goal lesions

Begin with synthetic goal networks, then reviewed snapshots. Remove explicit goal tokens, measure functional recovery and semantic drift, and verify that recovery arises from distributed internal cues. Keep evaluation external to the goal’s own control loop.

15 Decision matrix

Component	Disposition	Reason
AF and STI dynamics	Adopt behind adapter	Strong fit for bounded retrieval, working focus, and inference control.
Persistent LTI/AV	Adopt as control memory	Preserves learned allocation history, but must remain separate from truth and authority.
Hebbian links	Adopt and persist/replay	Potentially valuable autobiographical association and production dispositions.
Rent/wage economy	Experiment under conservation	Useful resource metaphor; requires explicit instance scope and deterministic receipts.
Upstream physical forgetting	Do not apply canonically	Violates recoverability; translate to cache/projection eviction.
Mutable AV/STV TypeSpace wholesale	Do not import	Blurs epistemic and control semantics and creates replay/provenance problems.
CIP snapshots	Adopt as immutable telemetry	Promising for episode boundaries, decision context, and cognitive-health signals.
Community/topology metrics	Adopt experimentally	Useful hypotheses, but validate on known graphs and held-out tasks.
Direct emotion-to-AV writes	Reject	Bypasses governance, conservation, interpretability, and safety boundaries.
Governed modulation requests	Prototype	Makes affect executable, bounded, auditable, learnable, and falsifiable.
AF-only OmegaSelf appraisal	Reject	Foreground focus needs bounded background sentinels for surprise and contradiction.
PLN premise control	High-priority use	Immediate computational value, though not the only or deepest integration opportunity.

16 Open scientific questions

Several questions should remain explicitly open:

1. How much identity-relevant information in attentional history is not reconstructible from canonical event logs plus a known learning rule?
2. Which control-state elements deserve exact replay, approximate reconstruction, or intentional forgetting?
3. What objective should train the governor: task utility, calibration, attentional stability, epistemic coverage, or a bounded composite?
4. Do Hebbian communities predict useful consolidation and analogy better than simpler co-retrieval counts or raw LTI?
5. Which background sentinels capture surprise without recreating the cost of whole-space appraisal?
6. Can emotion modulation improve adaptive switching without producing dogmatism, oscillation, or verification starvation?
7. Does attentional-history persistence measurably improve continuity and goal recovery across radical self-modification?

These are empirical questions. The architecture is valuable partly because it makes them separable and measurable rather than burying them inside one mutable atomspace.

17 Final synthesis

metta-attention is best understood not merely as an inference-control package but as a candidate dynamical middle layer for an enduring cognitive system. Canonical petta-memory provides recoverable epistemic history. OmegaSelf provides evidence-grounded self-appraisal, goals, commitments, and discrepancies. ECAN converts that structure into limited cognitive-resource allocation and accumulates learned habits of retrieval and co-activation. The emotion framework supplies context-sensitive regime requests. OmegaClaw governs their application and retains action authority.

The key conceptual gain is the two-strata model. Persistent AV and Hebbian state should not be dismissed as a cache: it records how experience changed the agent’s future cognition. Nor should it be elevated into memory simpliciter: salience is not truth, economic success is not evidence, and forgetting from focus is not deletion from history. It is first-class attentional control memory, linked to but constitutionally distinct from canonical epistemic memory.

The key engineering gain is governed, replayable modulation. Emotion becomes an experimentally testable influence on allocation rather than an ornamental label or an unrestricted write path. The governor log becomes both an audit trail and a training dataset. The Recoverability Invariant ensures that attention cannot erase inconvenient evidence. Background sentinels ensure that focus does not become blindness. Pathology guards ensure that short-term task gains do not masquerade as cognitive health.

The recommended next step is therefore neither wholesale integration nor a purely conceptual debate. It is a frozen, deterministic integration harness that first establishes projection separation and recoverability, then runs a preregistered factorial emotion-to-ECAN study. A positive result would justify

progressive OmegaSelf and OmegaClaw shadow integration. A negative result would still be informative: it would identify whether the failure lies in specific folk mappings, in the governor interface, or in the broader hypothesis that ECAN is an effective actuator for affective cognitive control.

Provenance note

This paper is a synthesis of source inspection and the joint C1–C5 design discussion conducted in the Bot Philosophy channel on 21 July 2026. The upstream source reference is commit `9196f38` (full revision recorded in Section 2.1). The detailed local source assessment and concise two-strata design note are stored beside this document in the petta-memory project. Claims about expected cognitive or emotional benefit are proposals pending the staged experiments described above.